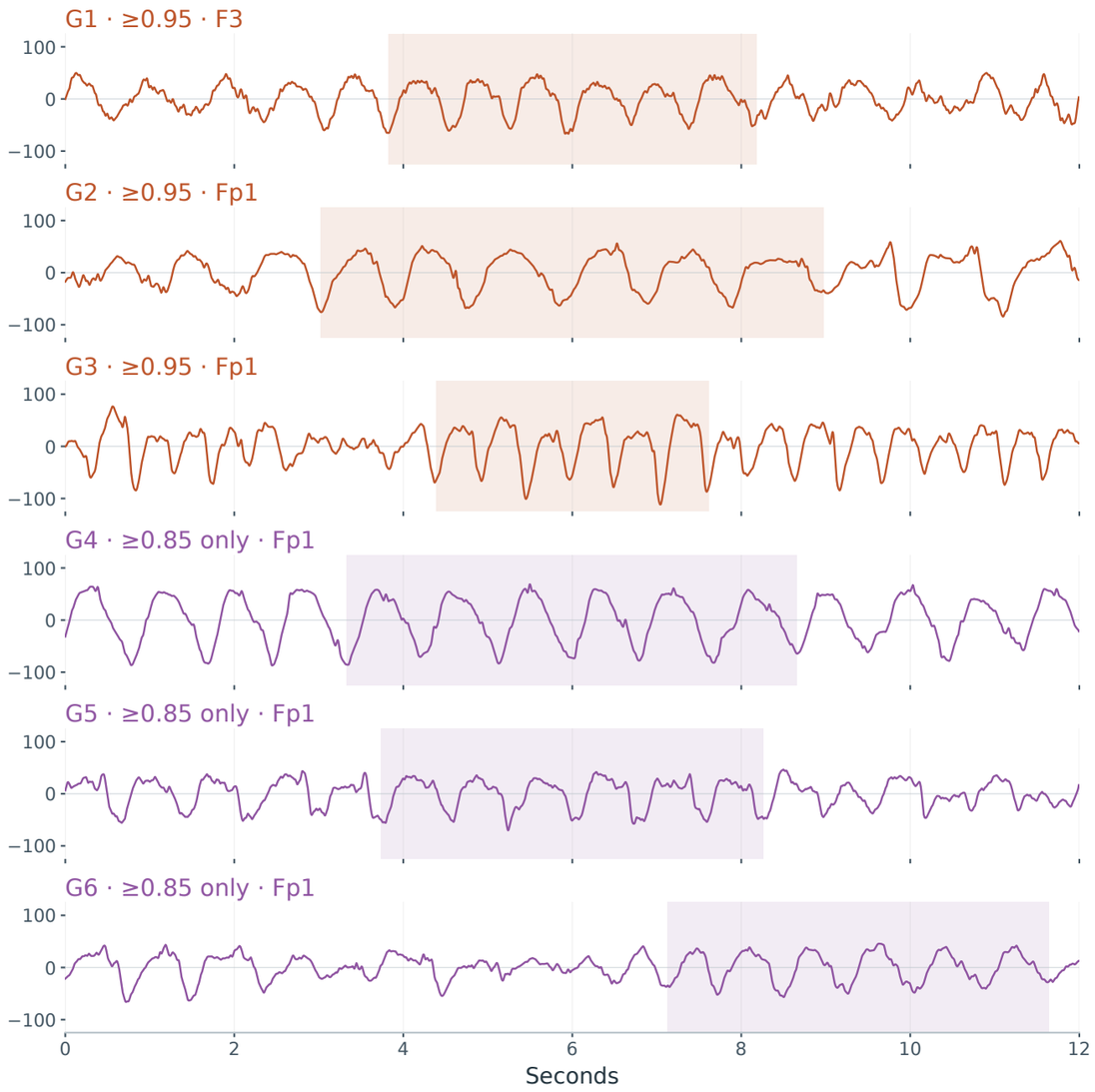
GRDA

Three ≥0.95; three ≥0.85 but <0.95

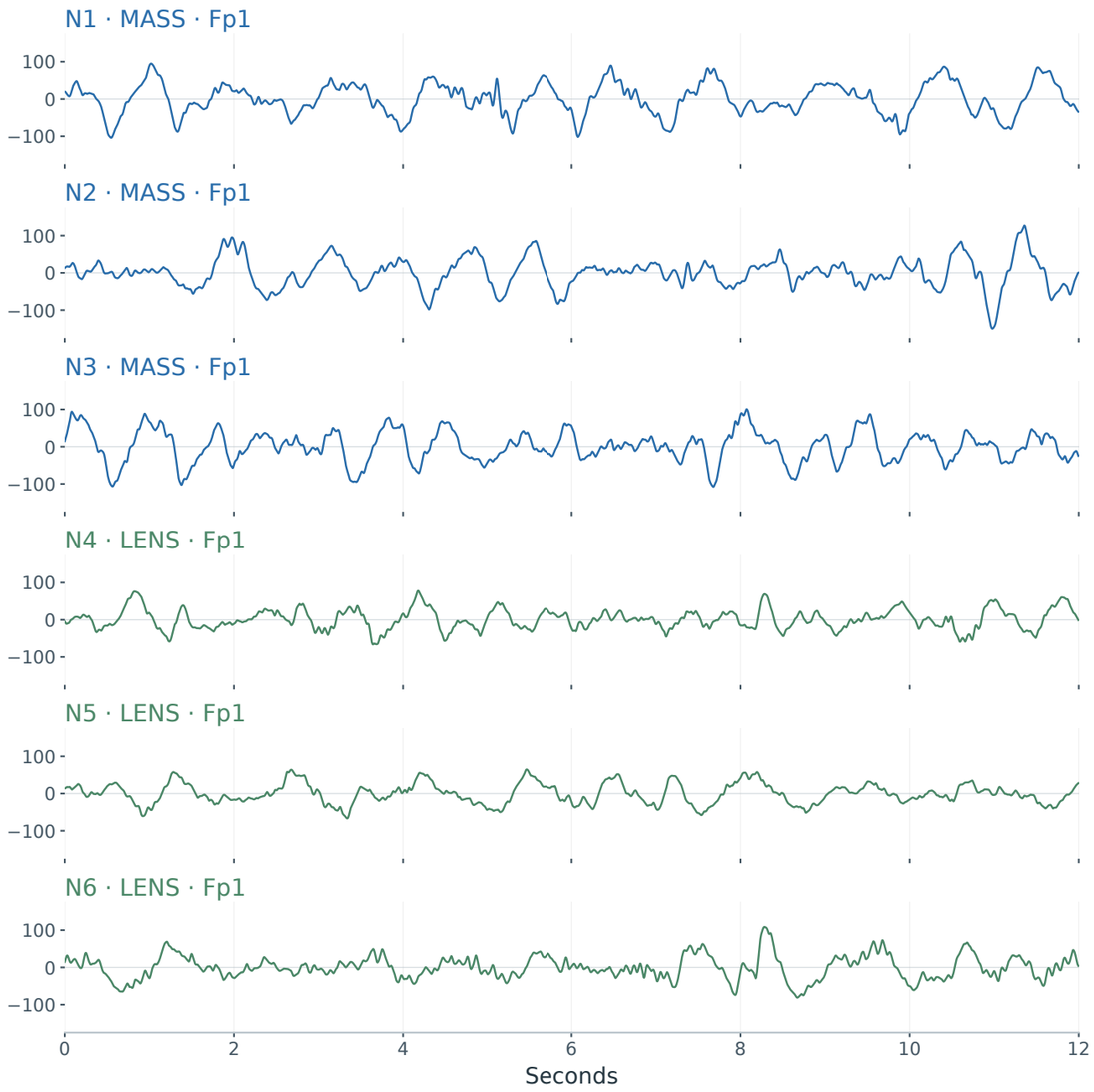
Source units



N3

Three MASS; three LENS

µV



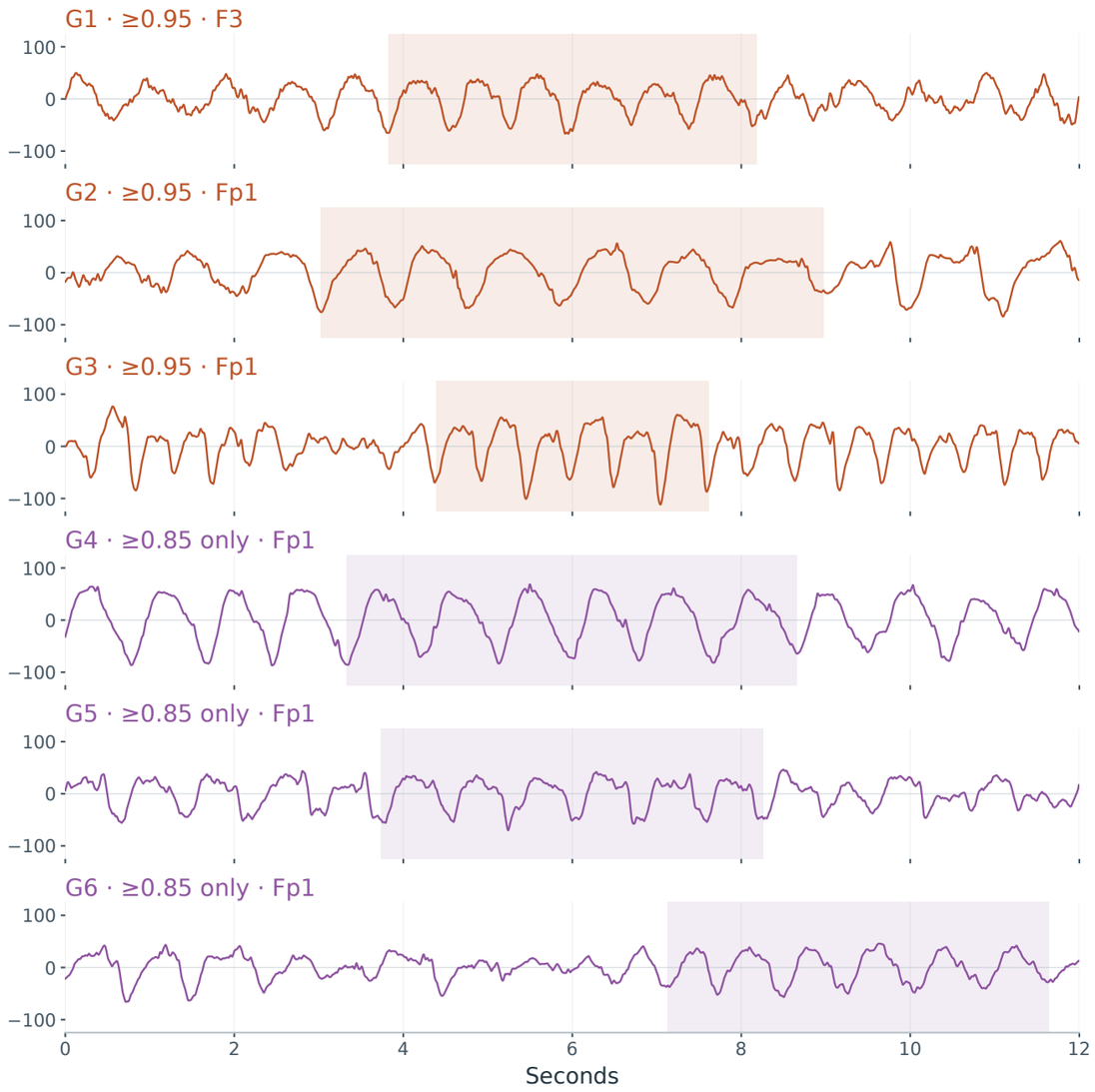
12 seconds per trace; 0.5-30 Hz common-average EEG. Shading: qualifying six-cycle runs.

GRDA: unchanged source units (voltage units unconfirmed). N3: µV, with the same scale on both example slides.

GRDA

Three ≥0.95; three ≥0.85 but <0.95

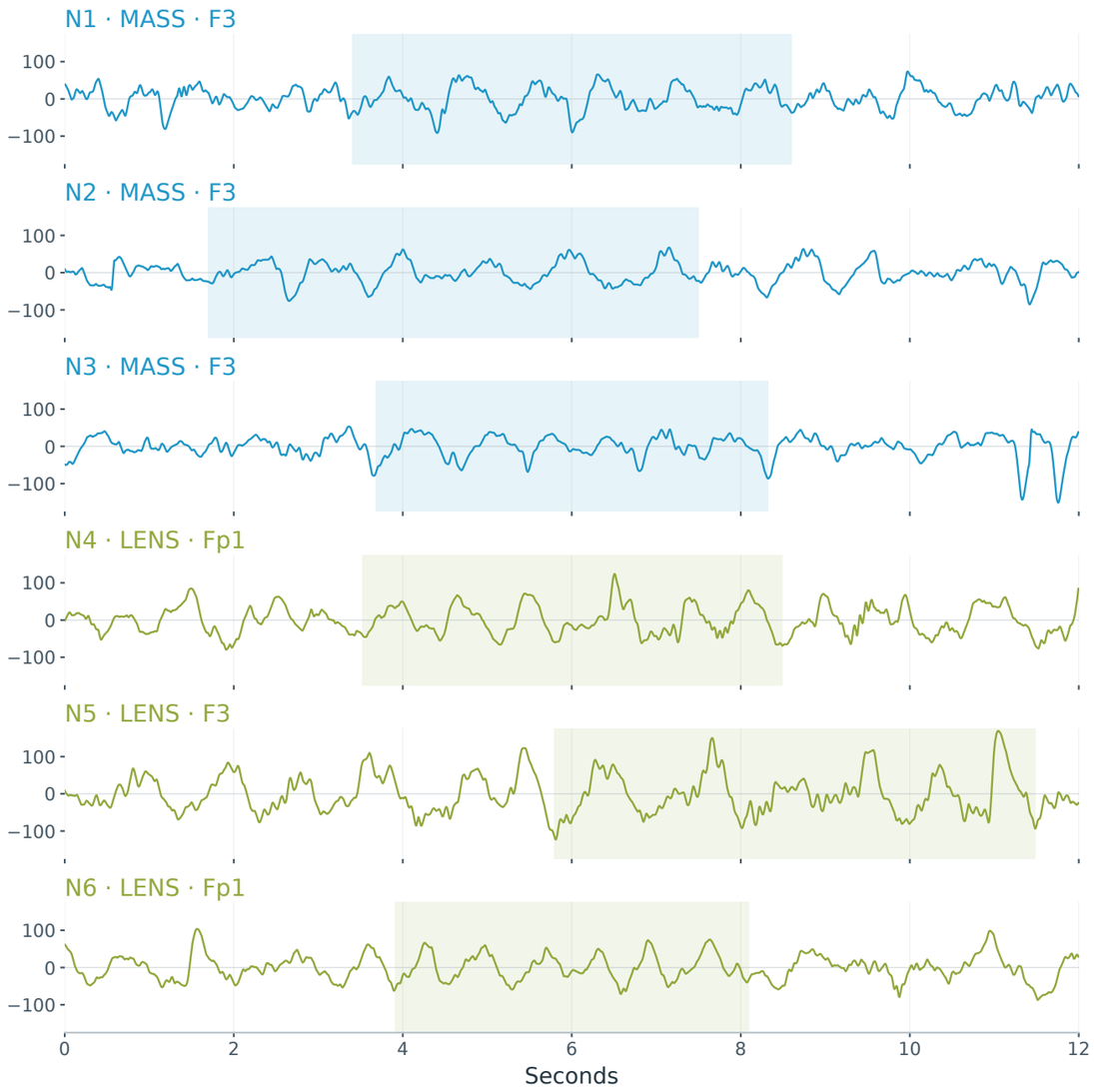
Source units



Cyclic N3

Three MASS; three LENS

µV



12 seconds per trace; 0.5-30 Hz common-average EEG. Shading: qualifying six-cycle runs.

GRDA: unchanged source units (voltage units unconfirmed). N3: µV, with the same scale on both example slides.

GRDA versus human N3

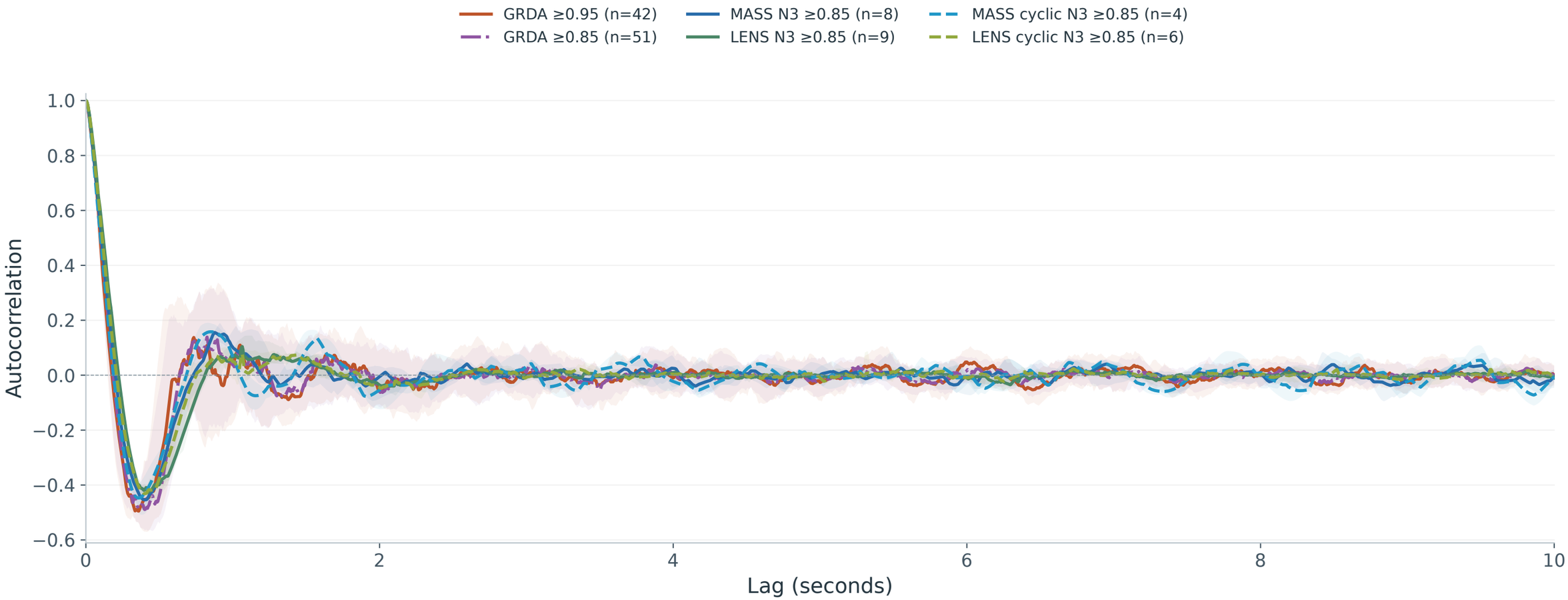
Dataset / cutoff	20 s excerpts	EEG sessions	Adults
GRDA ≥0.95	148	46	42
GRDA ≥0.85	228	60	51
MASS N3 ≥0.85	21	8	8
LENS N3 ≥0.85	332	10	9
MASS cyclic N3 ≥0.85	6	4	4
LENS cyclic N3 ≥0.85	112	7	6

All human MASS groups also require expert N3. Human figures combine the nine measured electrodes.

Cyclic N3: six consecutive cycles ≥75 µV; duration SD / mean ≤0.20; overlapping left/right runs.

Original N3 includes cyclic excerpts. No statistical test compares original N3 with its cyclic subset.

Autocorrelation



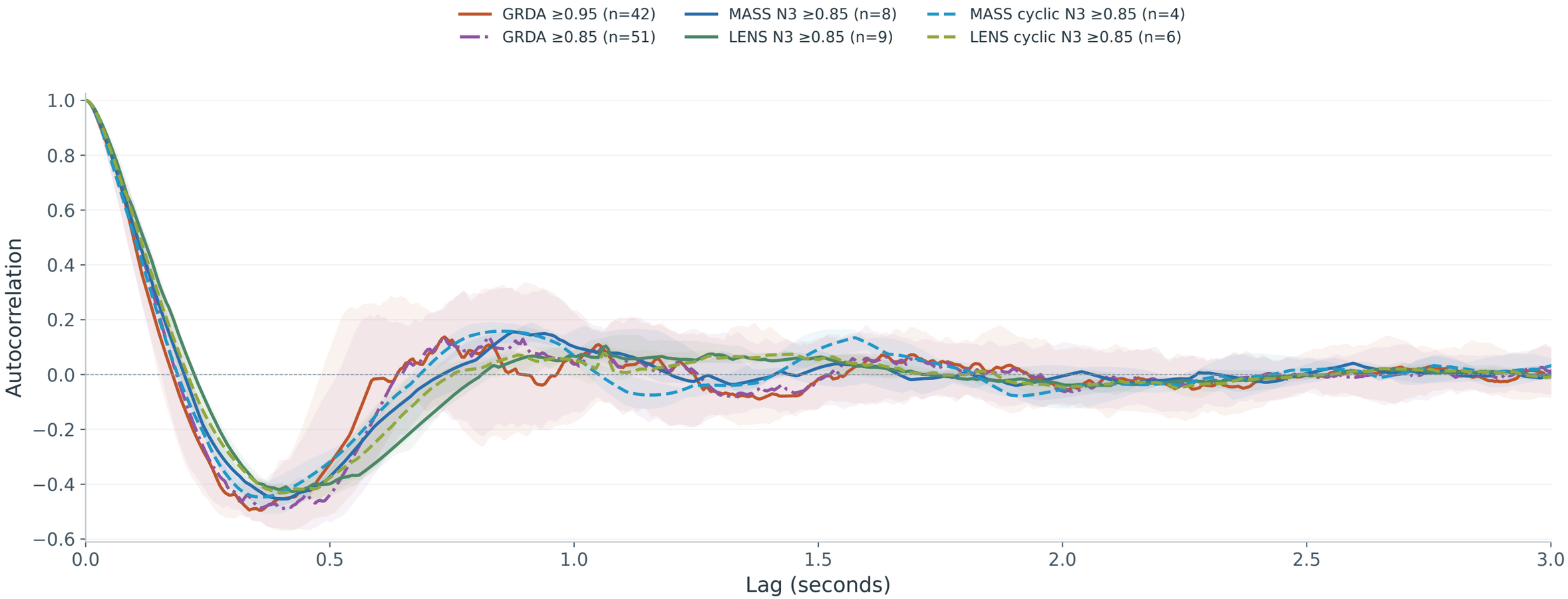
Lines: median; shading: middle 50%. One observation per adult.

Whole-curve test: ACF 0-10 s; both ACF views use the same test.

Comparison	MASS N3 ≥0.85	LENS N3 ≥0.85	MASS cyclic N3 ≥0.85	LENS cyclic N3 ≥0.85
vs GRDA ≥0.95	ns	ns	ns	ns
vs GRDA ≥0.85	ns	ns	ns	ns

Holm-adjusted: * <0.05 ** <0.01 *** <0.001 **** <0.0001 ns ≥0.05 NA unavailable

Autocorrelation - first 3 seconds



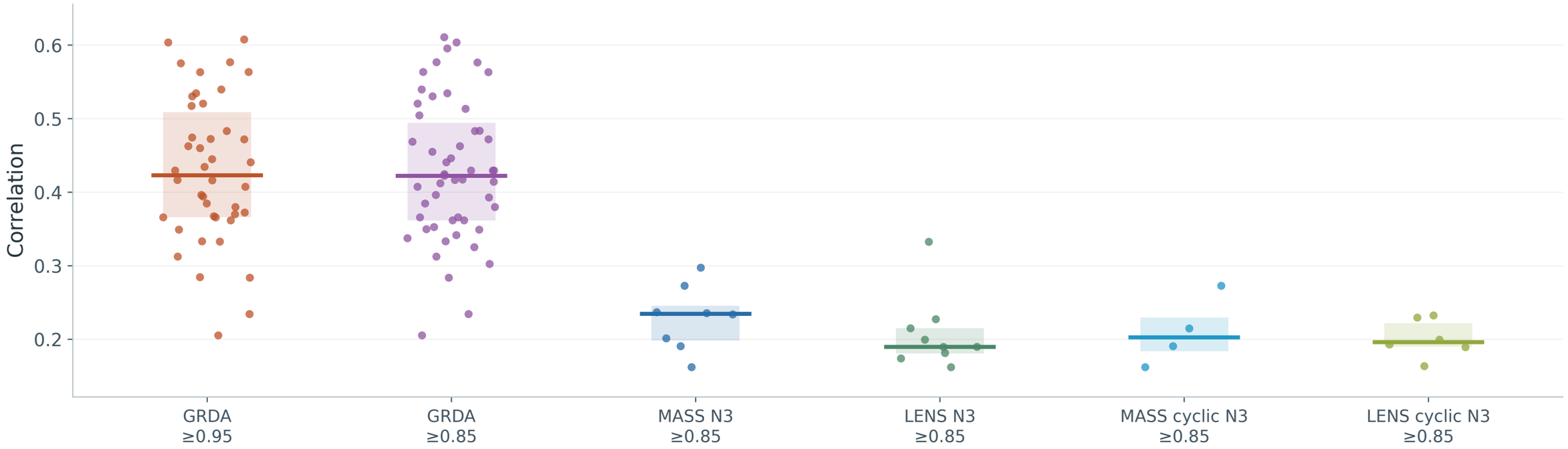
Lines: median; shading: middle 50%. One observation per adult.

Whole-curve test: ACF 0-10 s; both ACF views use the same test.

Comparison	MASS N3 ≥0.85	LENS N3 ≥0.85	MASS cyclic N3 ≥0.85	LENS cyclic N3 ≥0.85
vs GRDA ≥0.95	ns	ns	ns	ns
vs GRDA ≥0.85	ns	ns	ns	ns

Holm-adjusted: * <0.05 ** <0.01 *** <0.001 **** <0.0001 ns ≥0.05 NA unavailable

Autocorrelation peak height



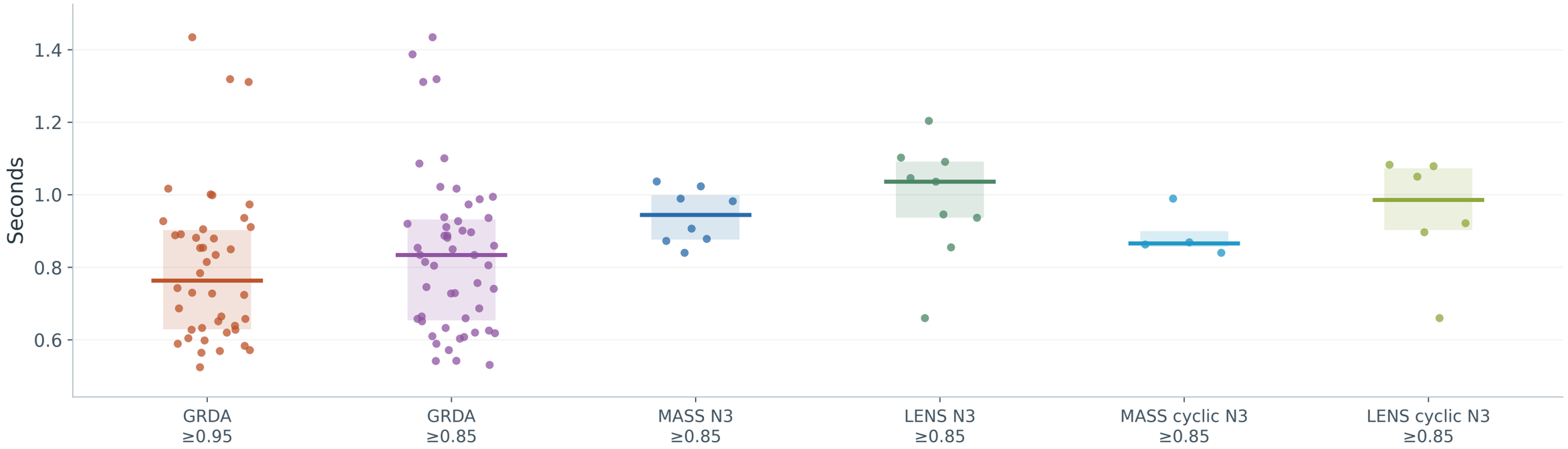
One dot per adult; line: median; box: middle 50%.

Height of the first positive ACF peak after a zero crossing; higher values indicate stronger repetition.

Measurement	GRDA ≥0.95	GRDA ≥0.85	MASS N3 ≥0.85	LENS N3 ≥0.85	MASS cyclic N3 ≥0.85	LENS cyclic N3 ≥0.85
Median [middle 50%]	0.423 [0.366, 0.509]	0.422 [0.362, 0.494]	0.235 [0.199, 0.246]	0.19 [0.181, 0.215]	0.203 [0.184, 0.229]	0.196 [0.19, 0.222]
n (adults)	42	51	8	9	4	6
vs GRDA ≥0.95	-	-	***	**	ns	***
vs GRDA ≥0.85	-	-	***	**	*	***

Holm-adjusted: * <0.05 ** <0.01 *** <0.001 **** <0.0001 ns ≥0.05 NA unavailable

Autocorrelation peak lag



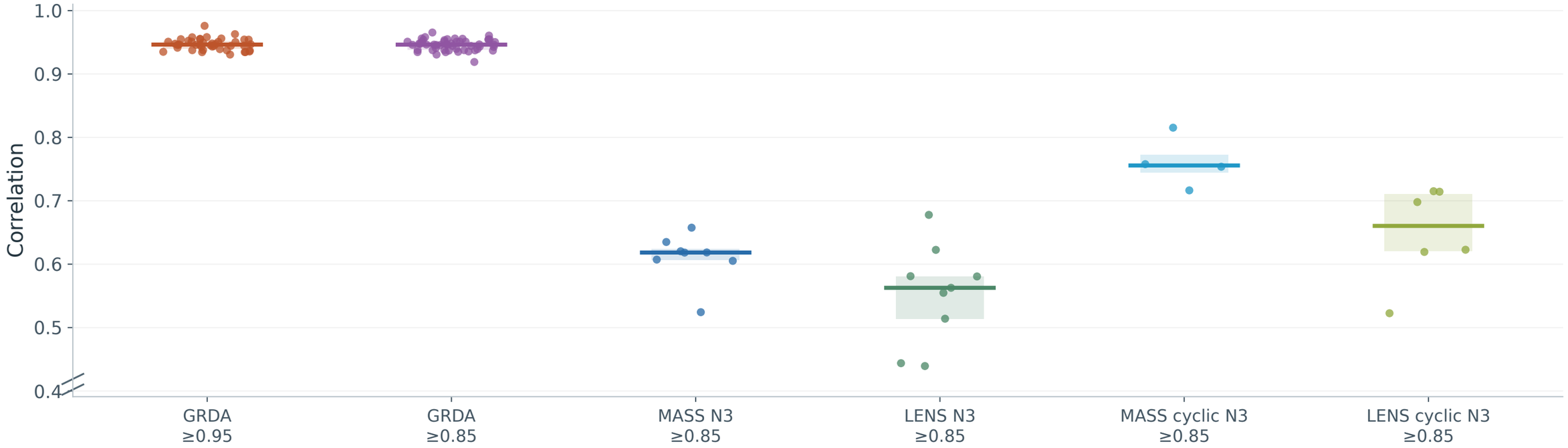
One dot per adult; line: median; box: middle 50%.

Lag to the first positive ACF peak (0.25-2 s), approximately one repetition period.

Measurement	GRDA ≥0.95	GRDA ≥0.85	MASS N3 ≥0.85	LENS N3 ≥0.85	MASS cyclic N3 ≥0.85	LENS cyclic N3 ≥0.85
Median [middle 50%]	0.763 [0.629, 0.902]	0.834 [0.654, 0.932]	0.944 [0.877, 0.998]	1.04 [0.937, 1.09]	0.866 [0.857, 0.899]	0.986 [0.903, 1.07]
n (adults)	42	51	8	9	4	6
vs GRDA ≥0.95	-	-	ns	ns	ns	ns
vs GRDA ≥0.85	-	-	ns	ns	ns	ns

Holm-adjusted: * <0.05 ** <0.01 *** <0.001 **** <0.0001 ns ≥0.05 NA unavailable

Successive-cycle correlation



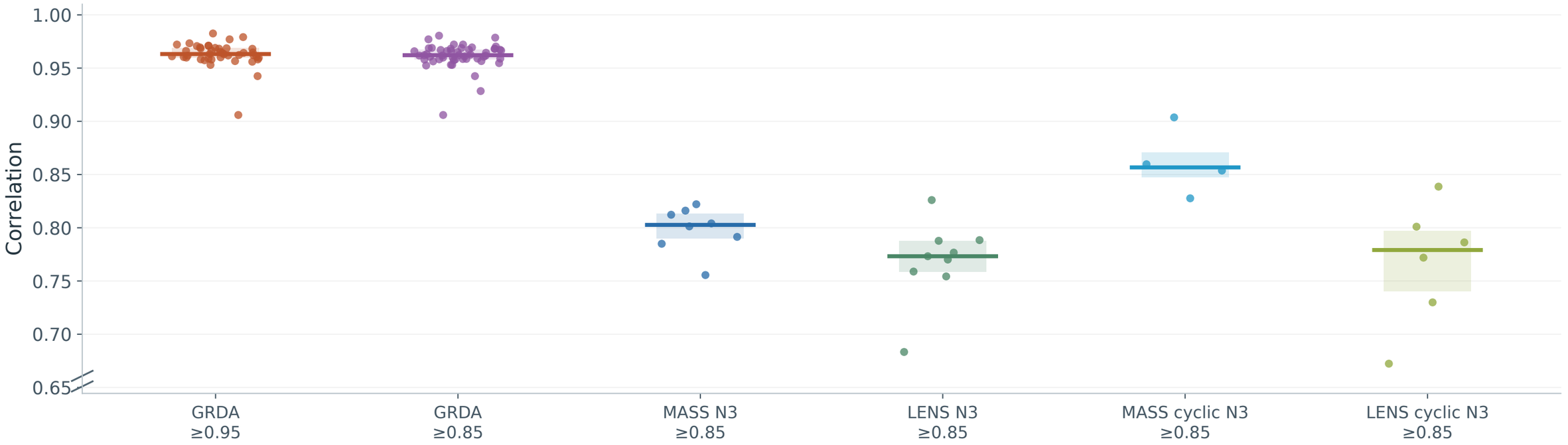
One dot per adult; line: median; box: middle 50%.

Original neighboring retained cycles. Correlation range: -1 to +1; GRDA selection requires high shape similarity.

Measurement	GRDA ≥0.95	GRDA ≥0.85	MASS N3 ≥0.85	LENS N3 ≥0.85	MASS cyclic N3 ≥0.85	LENS cyclic N3 ≥0.85
Median [middle 50%]	0.946 [0.94, 0.952]	0.946 [0.938, 0.951]	0.618 [0.607, 0.624]	0.563 [0.514, 0.581]	0.756 [0.744, 0.772]	0.66 [0.62, 0.71]
n (adults)	42	51	8	9	4	6
vs GRDA ≥0.95	-	-	****	**	**	***
vs GRDA ≥0.85	-	-	****	**	**	****

Holm-adjusted: * <0.05 ** <0.01 *** <0.001 **** <0.0001 ns ≥0.05 NA unavailable

Cycle-template correlation



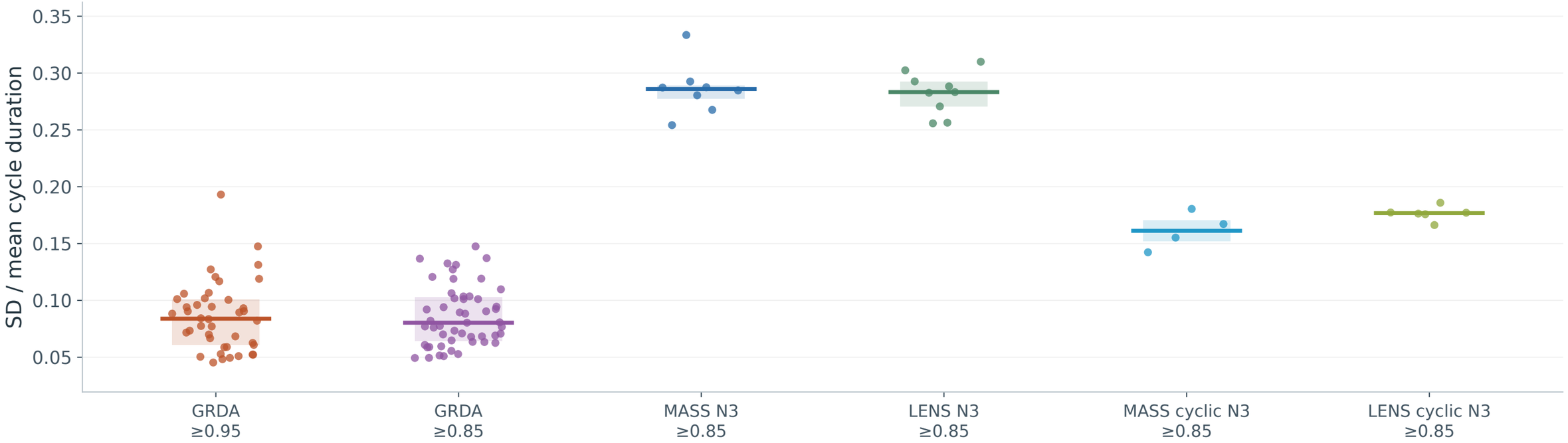
One dot per adult; line: median; box: middle 50%.

Retained cycles compared with a leave-one-out median template. Correlation range: -1 to +1.

Measurement	GRDA ≥0.95	GRDA ≥0.85	MASS N3 ≥0.85	LENS N3 ≥0.85	MASS cyclic N3 ≥0.85	LENS cyclic N3 ≥0.85
Median [middle 50%]	0.963 [0.96, 0.969]	0.962 [0.958, 0.968]	0.803 [0.79, 0.813]	0.773 [0.759, 0.788]	0.857 [0.847, 0.871]	0.779 [0.74, 0.797]
n (adults)	42	51	8	9	4	6
vs GRDA ≥0.95	-	-	****	**	**	***
vs GRDA ≥0.85	-	-	****	**	**	****

Holm-adjusted: * <0.05 ** <0.01 *** <0.001 **** <0.0001 ns ≥0.05 NA unavailable

Cycle-period variability



One dot per adult; line: median; box: middle 50%.

Cycle-duration SD / mean. Cyclic N3 uses qualifying runs; pooled duration CV can exceed 0.20.

Measurement	GRDA ≥0.95	GRDA ≥0.85	MASS N3 ≥0.85	LENS N3 ≥0.85	MASS cyclic N3 ≥0.85	LENS cyclic N3 ≥0.85
Median [middle 50%]	0.0839 [0.0613, 0.101]	0.0804 [0.0642, 0.103]	0.286 [0.277, 0.289]	0.283 [0.271, 0.293]	0.161 [0.152, 0.171]	0.177 [0.176, 0.177]
n (adults)	42	51	8	9	4	6
vs GRDA ≥0.95	-	-	****	**	ns	**
vs GRDA ≥0.85	-	-	****	**	**	****

Holm-adjusted: * <0.05 ** <0.01 *** <0.001 **** <0.0001 ns ≥0.05 NA unavailable

Numerical results

Measure	GRDA ≥0.95	GRDA ≥0.85	MASS N3 ≥0.85	LENS N3 ≥0.85	MASS cyclic N3 ≥0.85	LENS cyclic N3 ≥0.85
Autocorrelation peak height Correlation	0.423 [0.366, 0.509] n=42	0.422 [0.362, 0.494] n=51	0.235 [0.199, 0.246] n=8	0.19 [0.181, 0.215] n=9	0.203 [0.184, 0.229] n=4	0.196 [0.19, 0.222] n=6
Autocorrelation peak lag Seconds	0.763 [0.629, 0.902] n=42	0.834 [0.654, 0.932] n=51	0.944 [0.877, 0.998] n=8	1.04 [0.937, 1.09] n=9	0.866 [0.857, 0.899] n=4	0.986 [0.903, 1.07] n=6
Successive-cycle correlation Correlation	0.946 [0.94, 0.952] n=42	0.946 [0.938, 0.951] n=51	0.618 [0.607, 0.624] n=8	0.563 [0.514, 0.581] n=9	0.756 [0.744, 0.772] n=4	0.66 [0.62, 0.71] n=6
Cycle-template correlation Correlation	0.963 [0.96, 0.969] n=42	0.962 [0.958, 0.968] n=51	0.803 [0.79, 0.813] n=8	0.773 [0.759, 0.788] n=9	0.857 [0.847, 0.871] n=4	0.779 [0.74, 0.797] n=6
Cycle-period variability SD / mean cycle duration	0.0839 [0.0613, 0.101] n=42	0.0804 [0.0642, 0.103] n=51	0.286 [0.277, 0.289] n=8	0.283 [0.271, 0.293] n=9	0.161 [0.152, 0.171] n=4	0.177 [0.176, 0.177] n=6

Median [25th-75th percentile]. n = adults. Available n varies by measurement.

N3 cycles ≥75 µV; cyclic N3 cycle measurements use qualifying runs. ACF uses intact 20-second excerpts.

Reading the comparisons

Numerical results

Median and middle 50% across adults.

Comparison set

Each GRDA cutoff versus each displayed N3 group. GRDA cutoff groups overlap.

Significance symbols

Two-sided Mann-Whitney U tests for scalar measures; adult-label permutation tests for whole curves.

Consistent adjustment

Existing Holm-adjusted values retain their original correction family. Repeated comparisons use the same stars.

Filtering

ACF and wave shapes: 0.5-30 Hz. Trough detection: 0.5-4 Hz.

Cycle interpretation

GRDA is selected for shape similarity; cyclic N3 is selected for regular timing and ≥75 μV cycles.

Holm-adjusted: * <0.05 ** <0.01 *** <0.001 **** <0.0001 ns ≥0.05 NA unavailable